[INCH-POUND]
A-A-59130
17 October 1997

COMMERCIAL ITEM DESCRIPTION

ANALYZER, SPECTRUM
10 KHZ to 50 GHZ

The General Services Administration (GSA) has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE.

This CID outlines the minimum requirements for a programmable spectrum analyzer, with up to 100 Hertz (Hz) resolution bandwidth, which operates in the 10 Kilohertz (KHz) to 50 Gigahertz (GHz) range. The spectrum analyzer's features shall also include internal preselection and external mixers for electronic testing up to 50 GHz sweeps. Only those manufacturers that meet or surpass the requirements outlined in this CID are qualified. Test equipment offered shall be commercially available and may be modified to the extent necessary to meet the following description.

2. SALIENT CHARACTERISTICS.

2.1. Safety and Environmental. The analyzer shall be solid state and enclosed to provide protection from electrical shock. This spectrum analyzer may be used as bench top or rackmount equipment.

2.2. Electrical Power Sources. The analyzer shall operate from nominal commercial, military, and shipboard power sources of 110/220 volts AC (±10 percent) at line frequencies of 45 to 440 Hz. Power consumption shall not exceed 650 Volt Amps (VA). Transient state conditions shall meet MIL-PRF-28800F requirements for Class III equipment.

2.3. Temperature Requirements. In an operating environment of 0° to plus 55° Centigrade, the analyzer shall meet its performance and accuracy requirements. After storage in a non-operating

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environment of minus 40° to plus 65° Centigrade, the equipment shall meet its performance and accuracy requirements.

2.4. Preconditioning Requirement. Each deliverable analyzer shall be capable of being subjected to the environmental screening of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

2.5. Calibration and Maintenance Adjustments. Calibration adjustments shall be provided by the variable value components which are adjustable by the use of simple means. Any calibration by substitution of selected components or parts is unacceptable unless specifically approved in advance. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours.

2.6. Electromagnetic Interference (EMI). The analyzer shall comply with the following EMI requirements as specified:

2.6.1. CE01: Narrow Band Requirement from 10 to 15 KHz only.
2.6.2. CE03: Relaxed 20 dB from 15 KHz.
2.6.3. CS01: Full Limits
2.6.4. CS02: Full Limits
2.6.5. CS06: Full limits
2.6.6. RE01: Relaxed 15 dB at the fundamental through the 10^{th} harmonic of the source power frequency and exception 30 to 50 KHz.

2.7. Mean Time Between Failures (MTBF). The MTBF of this equipment shall be 2,000 hours. The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours.

3.0 PERFORMANCE CHARACTERISTICS.

Unless otherwise specified, all vernier controls shall be in the calibrated position and all performance requirements shall apply, with or without probes, after a one-hour warm-up period in a laboratory ambient operating temperature.

3.1. Frequency Range. The required frequency range of the spectrum analyzer with internal and external mixers shall be 10 KHz to 50 GHz.

3.1.1. Frequency Reference Accuracy.

3.1.1.1. Aging rate after 24-hour warm-up period, shall be less than 1 x 10 m9 per day, and less than 2.5 x 10 m7 per year.

3.1.1.2. Temperature Stability shall be better than 2 x 10 m8 for any temperature change between 0° and plus 55° Centigrade.

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3.2. Center Frequency Accuracy. Center frequency accuracy at 1 GHz with zero span shall be better than ± 50 Hz per day; span per division controls shall not affect accuracy more than ± 2% for spans less than 1 MHz.

3.3. Frequency Counter. The analyzer shall provide a fully integrated frequency counter capability. The frequency counter's accuracy shall allow a 2 GHz signal to be measured within 400 Hz. The total accuracy shall be defined as: ± counter frequency x reference accuracy plus (50 Hz x harmonic mixing number) plus 1 LSD.

3.4. Frequency Span. The minimum span range shall be 100 Hz to a frequency range of 21 GHz over 10 divisions CRT horizontal axis. The frequency span shall be better than ±10 percent over all spans.

3.5. Resolution Bandwidth. The resolution bandwidth shall be 10 Hz to 3 MHz (3 to 6 dB). The setting shall be front panel selectable or automatic.

3.6. Residual Frequency Modulation. The total residual frequency modulation (FM) shall be less than 10 Hz peak to peak in 20 milliseconds, using a 10 Hz resolution bandwidth in internal mixing for span settings less than or equal to 100 KHz in the phase lock mode.

3.7. Frequency Drift. The frequency drift after a one-hour warm-up period, shall be less than 50 Hz per minute for a frequency span setting less than or equal to 100 KHz.

3.8. Sensitivity. The following shall be a minimum acceptable overall sensitivity at 10 Hz resolution bandwidth and lowest harmonic mixing mode: (This clause does not refer to individual frequency bands.)

3.8.1. Sensitivity with internal mixer:

Frequency Range	Sensitivity (dBm)
10 KHz to 50 KHz	- 95
50 KHz to 1.8 GHz	-110
1.8 GHz to 6.4 GHz	-122
6.4 GHz to 21 GHz	-102

3.8.2. Sensitivity with external mixer:

Frequency Range	Sensitivity (dBm)
18 to 26.5 GHz	-110
26.5 to 40 GHz	-108
33 to 50 GHz	-102

3.9. Frequency Response. When tuned to the specified frequency range, with attenuator set at 10 dB, the analyzer shall have the following:

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3.9.1. Frequency Response with internal mixer:

Frequency Range	Response dB max.
10 KHz to 1.8 GHz	± 1.5
1.8 GHz to 6.4 GHz	± 2.5
6.4 GHz to 13 GHz	± 3.5
13 GHz to 18 GHz	± 4.0
18 GHz to 21 GHz	± 5.0

3.9.2. Frequency Response with external mixer:

Frequency Range	Response dB max
21 GHz to 26.5 GHz	± 2.5
26.5 GHz to 50 GHz	± 2.5

3.10. Sideband Noise. The sideband noise shall be minus 86 dBc/Hz or less at 10 KHz offset from center frequency and in the lowest harmonic mixing mode.

3.11. Single Input Range. The minimum amplitude reference level range shall be from minus 115 dBm to plus 30 dBm with a safe input of +30 dBm in the log mode one Watt in the linear mode.

3.12. Input Attenuator Accuracy. The input attenuator shall have a minimum range of 0 to 60 dB with front panel selectable steps of 10 dB attenuator. The attenuator accuracy shall not exceed 3.0 dB over 60.0 dB range from 0 to 18 GHz, and 6.0 dB over the .60.0 dB range overall from 18 to 21 GHz.

3.13. RF Input Connector. For frequencies up to 21 GHz, the RF input connector shall be compatible with the Type N connectors. For frequencies from 21 to 26.5 GHz, the RF input connector shall be made through a combination of waveguide mixer(s), adapter(s), and connectors which are compatible and mate nondestructively.

3.14. Input Impedance. Input impedance shall be 50 ohms nominal and Voltage Standing Wave Ratio (VSWR) shall not be greater 3:1 up to 18 GHz; nor greater than 3.5:1 from 18 to 21 GHz.

3.15. Input Power. The analyzer shall accept alternating signals that are less than one Watt (+30 dBm) continuous over the frequency range of 10 KHz to 21 GHz. The waveguide input over the frequency range of 18 to 50 GHz, shall accept alternating signals that are less than or equal to 100 mW (+20 dBm) continuous.

3.16. External Reference Input. The spectrum analyzer shall be provided with an input to accommodate a 10 MHz external reference.

3.17. Analyzer Output. The analyzer's output shall be provided with an internal calibrator signal output to provide performance verification as determined by the manufacturer. The analyzer shall also include an X-Y recorder output, a first local oscillator output greater than +5 dBm and 50 ohm nominal impedance; a pen-lift control, and an IF output at -5 dBm level and 50 ohm nominal impedance.

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3.18. Vertical Scaling. Linear or log vertical scaling shall be selectable from the front panel. The log scale shall have a minimum of two selectable calibrated scales: 2dB per division and 10 dB per division.

3.19. Display Accuracy. The maximum allowable cumulative error in the log display mode shall be ± 4.0 db. The maximum accuracy at full scale display in the linear mode shall be ± 5 percent.

3.20. Third Order Intermodulation Distortion shall be -70 dBc or less at 5 GHz with an input level to the mixer of -30 dBm for each tone.

3.21. Harmonic Distortion shall be -60 dBc or less to 2 GHz, and -100 dBc or less from 2 to 21 GHz with an input level to the mixer of -40 dBm.

3.22. Local Oscillator Emissions referenced to the input mixer shall be -80 dBm or lower with 10 dB RF attenuation.

3.23. Sweep Time. The sweep time shall be front panel controlled for manual or automatic mode. In the manual mode, the timing shall be front panel controlled for a minimum timing range of 20 microseconds to 5 seconds per division.

3.23.1. Sweep Time Accuracy. The sweep time accuracy shall be at least ± 10 percent.

3.24. Digital Storage. Digital storage shall be provided with selectable modes that save maximum signal values and noise-average spectral displays of two signals. The digital storage function shall be capable of storing and displaying at least 8 spectrums including the CRT readout measurement parameters. When a battery is required for the digital storage circuitry, the battery life shall be at least 12 months under the normal operating conditions.

3.25. Markers. The spectrum analyzer shall provide frequency and amplitude markers, using one marker for absolute measurements and two markers for relative measurements. Marker positioning shall provide a readout of frequency and amplitude of any point, or the difference in amplitude or frequency between two points within the frequency range of the instrument.

3.26. CRT Viewing Area. The spectrum analyzer shall provide a minimum viewing area of 10 divisions wide by 8 divisions high, with each division greater than .7 centimeters. The graticule will include a log scale of 10, 5, 2, and 1 dB/division and a linear scale of 10 percent of reference level/division.

3.27. CRT Readout. The CRT shall be capable of displaying the following readout:
3.27.1. Reference level.
3.27.2. Center frequency.
3.27.3. Vertical display mode.
3.27.4. Frequency Span per division or frequency span.
3.27.5. Resolution bandwidth.

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3.27.6. RF attenuation.

3.28. Mixers. The spectrum analyzer shall include three mixers and, if required, a unit driving amplifier. The ranges and conversion loss of the mixers shall be as follows:

Mixer Frequency Range	Average Conversion Loss Maximum
18 to 26.5 GHz	-30 dB
26.5 to 40 GHz	-30 dB
33 to 50 GHz	-30 dB

3.29. IEEE Interface. The analyzer shall include an IEEE standard 488 interface function and connector. This will enable remote control of all front panel functions except the power switch, CRT intensity, frequency calibration, level calibration, and trigger level adjustment knob.

3.30. Programmability. The spectrum analyzer shall be fully programmable except for CRT display positioning, CRT visual controls, manual scan, level calibration and power.

3.31. Front Panel Controls. The following shall be front panel controllable:
3.31.1. Store display.
3.31.2. Erase display.
3.31.3. Operation in a non-storage display mode.
3.31.4. Storage and display of two traces, one active and one stored spectrum.
3.31.5. Normalization or trace substitution.
3.31.6. Peak hold.
3.31.7. Selective display and blanking.
3.31.8. Internal preselection.
3.31.9. Digital averaging.
3.31.10. Vertical display.
3.31.11. Linear display.
3.31.12. Logarithmic display.
3.31.13. Intensity.

3.32. Signal Identification. The analyzer shall have the capability, when external mixers are in use, of identifying a true signal from a false signal when preselection is not available. This capability may be selected from a front panel control or from menu selection.

3.33. Accessories. The spectrum analyzer shall be provided with at least, but not limited to, the following accessories:

3.33.1. Spare fuses.
3.33.2. Power cable, 115 Volts.
3.33.3. Operator manual.
3.33.4. Programmer manual.
3.33.5. Mixer - 18 to 26.5 GHz (if needed).

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3.33.6. Mixer - 26.5 to 40 GHz.
3.33.7. Mixer - 33 to 50 GHz.
3.33.8. Mixer amplifier (if required).
3.33.9. 50 ohm coaxial cable, 6 ft., with N to N connector.
3.33.10. N male to BNC female adapter.
3.33.11. 50 ohm coaxial cable, BNC to BNC connector.
3.33.12. GPIB cable, 2 meter, double shielded.

3.34. Dimensions and Weight. The size and weight of the spectrum analyzer shall be consistent with commercial practices and the following specifications:

3.34.1. The maximum dimensions of the analyzer shall be approximately 7 inches high, 13 inches wide and 19 inches deep.

3.34.2. The weight of the analyzer, excluding the operator and programming manuals, shall not exceed 50 pounds.

4. REGULATORY REQUIREMENTS.

In accordance with Section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

5. PACKAGING.

The items shall be packed in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping shall be in compliance with national motor freight classification and uniform freight classification. Shipping container shall be marked as specified in the contract or order.

6. QUALITY ASSURANCE.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID, and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery, and thereafter, as may be otherwise provided for under the provisions of the contract. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples. (The federal government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract).

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7. NOTES.

7.1. Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product manufactured to inch-pound dimensions exceeds the tolerances specified in the metric units, a request should be made to the contracting officer to determine if the product is acceptable.

7.2. Source of Documents. Federal and military specifications and handbooks are available from the Defense Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

7.3. Known Acceptable Products. The following products are known to have previously met the salient characteristics of this CID and these manufacturers need not resubmit bid samples for certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification.

Manufacturer	Model	NSN
Tektronix, Inc.	2794A1	6625-01-377-7903
Hewlett-Packard Co.	8563A-E36	6625-01-392-5724
Hewlett-Packard Co.	8563E-E50	NSL

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - MI

Preparing Activity:
Air Force - 82

Review Activity:
DLA - CC

Agent Activity:
Air Force - 99

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3. DOCUMENT TITLE
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4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

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